MODULE-16

CLOUD FUNDAMENTALS

Cloud Computing

the cloud is a network of remote servers around the world that store data, run applications, and power services, so you don’t have to.

Cloud computing is all about accessing the resources over the internet.

- Infrastructure is provided over the internet by a cloud provider.
- Pay only for what you use (pay-as-you-go)
- Easily scalable on demand
- Provider handles maintenance and updates

Virtualisation

Virtualization is the process of creating multiple Virtual Machines (VMs) on a single physical computer using a Hypervisor.

- Better hardware utilization
- Cost reduction
- Easy scalability
- Foundation of cloud computing

Service-Oriented Architecture (SOA)

SOA is a software architecture where an application is divided into independent, reusable services that communicate over a network.

- Reusable services
- Loose coupling
- Easy integration
- Platform independent

Cloud Service Models

1. **IaaS (Infrastructure as a Service):** Provides virtual servers, storage, and networking. Users manage the OS and applications.
    - Examples: AWS EC2, Azure Virtual Machines, Google Compute Engine
    - Used for: Hosting servers and applications.
2. **PaaS (Platform as a Service):** Provides a platform to develop, test, and deploy applications. Infrastructure is managed by the provider.
    - Examples: Google App Engine, Azure App Service, AWS Elastic Beanstalk
    - Used for: Application development.
3. **SaaS (Software as a Service):** Ready-to-use software delivered over the internet.
    - Examples: Gmail, Google Docs, Microsoft 365, Zoom
    - Used for: Daily business and productivity applications.

Cloud Deployment Models

1. **Public Cloud**
    - Owned by a cloud provider and shared among multiple users.
    - Examples: AWS, Azure, GCP
2. **Private Cloud**
    - Dedicated to a single organization for better security and control.
    - Example: A bank's internal cloud.
3. **Hybrid Cloud**
    - Combination of public and private clouds.
    - Example: Sensitive data in private cloud, website in public cloud.
4. **Community Cloud**
    - Shared by organizations with similar requirements.
    - Example: Multiple universities sharing research infrastructure.

Cloud Service Providers

1. Amazon Web Service:
    - Largest cloud provider.
    - Popular services: EC2, S3, RDS.
    - Best for general cloud solutions.
2. Microsoft Azure:
    - Microsoft's cloud platform.
    - Popular services: Azure VM, Blob Storage, App Service.
    - Best for organizations using Microsoft products.

3. Google Cloud Platform (GCP):
    - Google's cloud platform.
    - Popular services: Compute Engine, BigQuery, Cloud Storage.
    - Best for AI, Machine Learning, and Data Analytics.

Advantages:

- Cost Efficiency
- Scalability
- Performance enhancement
- Security and accessibility
- disaster recovery

Amazon EC2 (Elastic Compute Cloud)

Amazon EC2 (Elastic Compute Cloud) is an AWS service that provides virtual servers (instances) in the cloud to run applications.

- Resizable virtual machines
- Pay-as-you-go pricing
- Scalable
- Secure and highly available

Steps:

1. Sign in to the AWS Management Console.
2. Go to EC2 Dashboard.
3. Click Launch Instance.
4. Enter an Instance Name.
5. Select an Amazon Machine Image (AMI).
6. Choose an Instance Type (e.g., t2.micro).
7. Create or select a Key Pair.
8. Configure Security Group.
9. Click Launch Instance.

Amazon Machine Images (AMIs)

An Amazon Machine Image (AMI) is a pre-configured template used to launch an EC2 instance. It contains:

- Operating System (Windows/Linux)

- Software
- Configuration settings

Selecting AMI: Choose an AMI based on your application need

Creating an AMI:

- Configure an EC2 instance.
- Create an image from the instance.
- Reuse it to launch identical instances.

EC2 Instance Types

Definition

Instance types define the **CPU, memory, storage, and networking capacity** of an EC2 instance.

Common Types

- **t2.micro** – Free Tier, suitable for small applications and learning.
- **t3.medium** – Medium workloads.
- **m5.large** – General-purpose workloads.
- **c5.large** – Compute-intensive applications.

t2.micro provides **1 vCPU and 1 GB RAM** and is commonly used for AWS Free Tier.

Security Group

A Security Group is a virtual firewall that controls traffic to and from an EC2 instance.

- Inbound Rules: Control incoming traffic (e.g., SSH, HTTP).
- Outbound Rules: Control outgoing traffic from the instance.

Key Pairs

A Key Pair is used to securely connect to an EC2 instance.

It consists of:

- Public Key – Stored by AWS.
- Private Key (.pem file) – Downloaded by the user and kept secure.

Steps

- Create a new key pair while launching the instance.
- Download the .pem file.

- Store it safely (cannot be downloaded again).
- Use the .pem file to connect to the EC2 instance via SSH.

Docker and Container Basics

A container is a lightweight, portable package that contains an application along with all its dependencies.

Containers are

- Lightweight
- Portable
- Isolated
- Fast startup

Docker: Docker is a platform used to create, run, and manage containers.

Amazon ECS

Amazon Elastic Container Service (ECS) is a managed AWS service for deploying and managing Docker containers.

- Deploy containerized applications
- Manage multiple containers
- Automatically scale containers

Unlike EC2, which is used for any application, EC2 can be only used for Docker applications.

Creating Containers in ECS

1. Create a Docker image.
2. Push the image to Amazon ECR.
3. Create an ECS Cluster.
4. Create a Task Definition.
5. Run the Task as a Service.

Amazon S3

Amazon Simple Storage Service (S3) is an object storage service used to store and retrieve files.

Bucket: Container that stores objects.

Object: A file stored inside a bucket.

Upload: Add files to the bucket.

Download: Retrieve files from the bucket.

Bucket Permissions:

- **Public:** Accessible by everyone
- **Private:** Accessible by authorized users.

Steps to create an S3 bucket:

1. Open S3 Console.
2. Click Create Bucket.
3. Enter bucket name and region.
4. Create the bucket.

Lifecycle Policy: Automatically moves or deletes objects after a specified period.

Versioning: Stores multiple versions of an object to recover deleted or modified files.

Amazon EBS (Elastic Block Store)

Amazon EBS provides persistent block storage for EC2 instances.

- Operating system storage
- Database storage
- Application data

Creating and Attaching an EBS Volume

1. Create an EBS volume.
2. Select Availability Zone.
3. Attach it to an EC2 instance.
4. Mount the volume.

Volume Types

- **gp2:** General Purpose SSD
- **gp3:** Improved General Purpose SSD
- **io1:** High-performance SSD

Snapshots: A snapshot is a backup of an EBS volume stored in Amazon S3.

Amazon VPC

A Virtual Private Cloud (VPC) is an isolated virtual network in AWS.

Two Types:

1. **Public Subnet:**
    - Internet accessible
    - used for web servers
2. **Private Subnet:**
    - No direct internet access
    - used for databases

Route Table: Determines how network traffic is routed.

Internet Gateway (IGW): Allows **public subnet** resources to access the internet.

NAT Gateway: Allows **private subnet** resources to access the internet without exposing them publicly.

Security Group: Acts as an instance-level virtual firewall.

VPC Peering: Connects two VPCs so they can communicate securely.

Elastic Load Balancer (ELB)

Elastic Load Balancer distributes incoming traffic across multiple servers.

1. **Application Load Balancer (ALB):**
    - Present on the Layer 7 of the OSI Model
    - Supports HTTP/HTTPS
    - Path-based routing
    - web applications

2. **Network Load Balancer (NLB):**
    - Present on the Layer-4 of the OSI Model
    - Supports TCP/UDP
    - High performance networking
    - high-performing applications

Target Group - A group of servers that receive traffic from the load balancer.

Health Check - Checks server health and routes traffic only to healthy servers.

Amazon RDS

Amazon Relational Database Service (RDS) is a managed relational database service.

Through RDS, AWS manages maintenance and Automates backups.

Multi-AZ Deployment: Creates a standby database in another Availability Zone for high availability and automatic failover.

Automated Backups: AWS automatically backs up the database.

Snapshots: Manual backups that can be restored later.

Amazon DynamoDB

Amazon DynamoDB is a fully managed NoSQL database for key-value and document data.

RDS	DynamoDB
Relational (SQL)	NoSQL
Fixed schema	Flexible schema
Complex queries	Fast key-based access

Primary Keys

- **Partition Key:** Uniquely identifies an item.
- **Sort Key:** Sorts items sharing the same partition key.

Query vs Scan

- **Query:** Searches using the primary key (fast).
- **Scan:** Reads the entire table (slow).

AWS LAMBDA

AWS Lambda is a serverless compute service that executes code automatically in response to events.

Serverless Computing: Running code without managing servers.

Event Triggers:

- Amazon S3
- API Gateway
- DynamoDB Streams

Pricing: Pay only for the number of executions and execution time.

Integration: Lambda integrates with:

- Amazon S3
- API Gateway
- DynamoDB
- SNS
- SQS

AWS API Gateway

AWS API Gateway is a service used to create, publish, secure, and manage APIs.

Creating a REST API

1. Open API Gateway.
2. Create a REST API.
3. Add routes and HTTP methods.
4. Integrate with Lambda.
5. Deploy the API.

HTTP Methods

- GET – Retrieve data
- POST – Create data
- PUT – Update data
- DELETE – Delete data

Routes: Maps a URL path to a backend service.

Example: /users → Lambda Function

Deployment Stages

- Development (Dev)
- Staging
- Production (Prod)

Throttling: Limits the number of API requests to prevent overload.

API Security

- IAM Authentication
- API Keys
- AWS Cognito
- Lambda Authorizers